Irfan Ullah

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LinkedIn | Portfolio | Open to relocation

Research Profile

Final-year Ph.D. researcher in Electrical Engineering specializing in computational power-system modeling, optimization, and AI-assisted decision methods for resilient electric power systems. Five years of experience developing optimization models, research software, and automated simulation workflows using Python, Julia/JuMP, Gurobi, PowerModelsONM.jl, PowerModelsDistribution.jl, and OpenDSS. Research spans mixed-integer and robust optimization, decomposition-based methods, rolling-horizon decision making, dynamic programming, DER/BESS coordination, uncertainty-aware simulation, and optimization-generated datasets for AI models. Completed two summer research internships at Los Alamos National Laboratory and participated in research collaboration with Purdue University on decision-transformer-based grid restoration.

Technical Expertise

Optimization and OR:	Mixed-integer linear programming; two-stage robust optimization; master–subproblem decomposition; rolling-horizon optimization; multi-agent dynamic programming; uncertainty sets; scenario analysis; convex and linearized power-flow formulations; Gurobi; Ipopt.
Energy Systems:	Multi-phase unbalanced distribution systems; optimal power flow; maximal load delivery; DER and BESS integration; feeder reconfiguration; microgrids; grid-forming resources; voltage regulation; contingency and restoration analysis.
Computational Methods:	Large-scale simulation experiments; scenario generation; optimization-generated datasets; model validation; experiment automation; visualization; performance analysis; reproducible research workflows.
Programming and Tools:	Python, Julia, JuMP, MATLAB/Simulink, Git/GitHub, Linux, PyQt; PowerModelsONM.jl, PowerModelsDistribution.jl, OpenDSS, PowerPlots, PSCAD, PSS/E, PowerWorld, CYME, ETAP.

Research Experience

Texas A&M University, Department of Electrical & Computer Engineering — **Aug. 2022–Present**
Graduate Research Assistant – Power & Energy Group — *College Station, TX*

- Developed and evaluated mixed-integer black-start restoration methods for large-scale, multi-phase distribution feeders, incorporating DER/BESS dispatch, grid-forming inverter constraints, voltage limits, radiality, switching decisions, cold-load pickup, and sequential load restoration.
- Implemented a two-stage robust optimization framework for maximal load delivery using master–subproblem decomposition, block-level load uncertainty, worst-case operating scenarios, and scenario-indexed validation on a modified Iowa 240-node distribution system.
- Developed rolling-horizon restoration frameworks with overlapping prediction and control windows, recursive backtracking, intertemporal storage coordination, and storage-capacity feasibility studies under changing operating conditions.
- Generated optimization-based restoration trajectories and contributed power-system formulations, data interpretation, and feasibility validation to decision-transformer and hybrid quantum physics-informed decision-transformer research conducted with Purdue University collaborators.
- Mentored undergraduate researchers in power-system modeling, simulation setup, data processing, and research-software workflow automation; prepared technical reports and presentations for collaborative grid-resilience research.

Los Alamos National Laboratory — **Summer 2025**
Summer Research Intern – Power Systems — *Los Alamos, NM*

- Collaborated with LANL staff scientists to evaluate operational strategies for resilient distribution systems using large-scale simulation and optimization models.
- Diagnosed power-flow inconsistencies, voltage-limit violations, cold-load-pickup implementation issues, and data-conversion errors in PowerModelsONM.jl restoration studies.
- Validated optimization solutions against OpenDSS simulations and traced discrepancies across input data, network models, formulations, and post-processing workflows; documented findings for research collaborators.

Los Alamos National Laboratory — **Summer 2024**
Summer Research Intern – Power Systems — *Los Alamos, NM*

- Developed uncertainty-aware load-block modeling approaches to evaluate the robustness of restoration and microgrid-reconfiguration decisions under demand variability.
- Implemented and evaluated overlapping rolling-horizon restoration workflows in PowerModelsONM.jl for large-scale distribution-system studies.
- Built Python workflows integrating OpenDSS, PowerModelsDistribution.jl, PowerModelsONM.jl, PowerPlots, and Excel for network preparation, batch simulations, scenario comparison, metric extraction, visualization, and engineering reporting.

Research Software and Platform Development

Founder & Lead Research Software Developer, GridResilience Studio **2025–Present**

Computational Platform for Power-System Optimization and Restoration

Python, Julia, JuMP, PowerModelsONM.jl, PowerModelsDistribution.jl, OpenDSS, PyQt, Gurobi, Ipopt

- Founded and developed a research software platform for distribution-system analysis, optimization, and resilience studies, integrating OpenDSS network models with Julia/JuMP-based power flow, optimal power flow, maximal load delivery, and black-start restoration workflows.
- Implemented steady-state and multi-time-step unbalanced power flow and OPF with voltage, thermal, ampacity, DER, BESS, transformer-tap, capacitor-control, and intertemporal storage constraints.
- Designed a multi-agent dynamic-programming restoration framework coordinating topology, priority-based load pickup, DER/BESS dispatch, and electrical-feasibility agents to determine sequential restoration policies.
- Enforced active/reactive power balance, radiality, voltage limits, branch apparent-power limits, source capability, grid-forming island requirements, switching limits, storage efficiency, and terminal energy-reserve constraints.
- Validated the restoration framework on a three-bus system through 20 automated tests, reproducing the optimal switching sequence, load-restoration trajectory, voltage profile, branch loading, and BESS dispatch.
- Created interactive system building, network visualization, scenario generation, time-step playback, result comparison, OpenDSS validation, and JSON/CSV/Excel reporting; added GridMate, an offline diagnostic assistant for solver, topology, voltage, thermal, and infeasibility analysis.

Selected Research Projects and Contributions

- **Sequential Black-Start Restoration Using PowerModelsONM.jl:** Implemented a full-horizon optimization framework coordinating DER dispatch, BESS operation, switching, radiality, voltage feasibility, and prioritized load restoration; developed automated feeder conversion, optimization, validation, and visualization workflows.
- **Two-Stage Robust Maximal Load Delivery:** Developed a decomposition-based robust optimization framework for load restoration under block-level demand uncertainty; evaluated robustness, load delivery, topology, and computational performance on a modified Iowa 240-node feeder.
- **Rolling-Horizon and Uncertainty-Aware Restoration:** Studied how prediction-horizon length, early battery dispatch, forecast error, and path-dependent switching affect future restoration feasibility; implemented recursive backtracking and sequential storage-capacity reinforcement.
- **AI-Assisted Distribution-System Restoration:** Generated optimization-based restoration trajectories, modeled uncertain load and renewable generation, incorporated physical constraints, and supported feasibility evaluation of learned decisions in decision-transformer research.
- **Converter-Based Power Quality and Energy Integration:** Modeled solid-state-transformer and UPQC architectures in MATLAB/Simulink, analyzing renewable integration, multi-port converter operation, DC-link behavior, voltage sag/swell mitigation, harmonic reduction, and fault-condition performance.
- **Data-Center Power and Energy Routing:** Studying multi-port solid-state-transformer architectures for resilient data-center power systems with renewable generation and BESS, including bidirectional AC/DC power routing, coordinated source/storage operation, power-quality support, and resilient energy management.

Selected Publications

1. **I. Ullah**, K. L. Butler-Purry, and H. Nagarajan, “Two-Stage Robust Optimization Maximal Load Delivery Framework on a Large-Scale Distribution System,” presented at the *IEEE Power & Energy Society General Meeting (PESGM)*, Montréal, Canada, July 20, 2026.
2. **I. Ullah**, K. L. Butler-Purry, R. Bent, H. Nagarajan, and A. Barnes, “Studying a Sequential Black Start Restoration Method Using PowerModelsONM.jl,” *North American Power Symposium (NAPS)*, Hartford, CT, USA, 2025.
3. **I. Ullah** and K. L. Butler-Purry, “Evaluation of a Black Start Distribution System Restoration Method Using a Declarative Modeling Framework,” submitted to *IEEE Transactions on Power Systems*, June 2026.
4. H. Zhao, J. Wei-Kocsis, **I. Ullah**, and K. L. Butler-Purry, “HQ-PIDT: A Hybrid Quantum Physics-Informed Decision Transformer for Resilient Distribution System Restoration Under Uncertainty,” submitted to *IEEE Transactions on Industrial Informatics*, July 2026.
5. W. Ahmad, L. Khan, Q. Khan, and **I. Ullah**, “Advanced Sliding Mode Control Techniques Employed on Electric Hybrid Vehicle,” *IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, 2024, pp. 1–6.
6. **I. Ullah**, S. Rahman, and I. A. Khan, “SST-Based Marine Shipboard System for Improved Performance and Renewable Integration,” *IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, 2023.

Education

Texas A&M University **2022–Present**

Ph.D. in Electrical Engineering, GPA: 3.5/4.0; Expected Spring 2027 College Station, TX Research: Distribution-system resilience, black-start restoration, robust optimization, DER/microgrid coordination, and AI-assisted grid decision methods.

Teaching and Mentoring

- **Teaching Assistant, ECEN 325, Texas A&M University, Spring 2026:** Supported laboratory and classroom instruction, student questions, assessment activities, and technical problem solving in electrical engineering.
- **Exam Grader, ECEN 214, Texas A&M University, two semesters:** Evaluated student solutions and applied consistent technical grading criteria for electrical-engineering examinations.
- **Undergraduate Research Mentor, Aggie STEM Research Program:** Guided students in distribution-system modeling, simulation setup, data analysis, research-software workflows, technical communication, and presentation preparation.

Selected Presentations

- "Two-Stage Robust Optimization Maximal Load Delivery Framework on a Large-Scale Distribution System," IEEE Power & Energy Society General Meeting, Montréal, Canada, July 20, 2026.
- "Studying a Sequential Black Start Restoration Method Using PowerModelsONM.jl," North American Power Symposium, Hartford, CT, 2025.
- "SST-Based Marine Shipboard System for Improved Performance and Renewable Integration," IEEE Texas Power and Energy Conference, College Station, TX, 2023.

Leadership, Service, and Awards

- **Session Chair; Communication and Promotion Chair, Texas Power & Energy Conference, 2024–2026:** Chaired technical sessions and coordinated communications, outreach, sponsor engagement, and industry-speaker activities.
- **Technical Reviewer, IEEE PES General Meeting and Texas Power & Energy Conference, 2023–2026:** Reviewed manuscripts in power systems, optimization, grid resilience, and distribution-system modeling.
- **Aggie STEM Research Program Leader:** Led and mentored undergraduate researchers in power-system modeling, simulation, data analysis, workflow development, and research presentations.
- **Prime Minister's National ICT Scholarship, 2012–2016:** Fully funded national scholarship recognizing academic achievement and leadership potential.
- **National ICT Initiative Research Award, 2015–2016:** Award and project funding for antenna design, simulation, prototyping, and experimental evaluation.

Professional References

Dr. Russell W. Bent
Los Alamos National Laboratory
Contact information available upon request

Dr. Harsha Nagarajan
Los Alamos National Laboratory
Contact information available upon request

Dr. Arthur Barnes
Los Alamos National Laboratory
Contact information available upon request